

FINAL REPORT: AI LECTURER PROJECT

Project Title:

AI Lecturer – Automated Course Content and Lecture Video Generation

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LETTER OF TRANSMITTAL

Dear Dr. Behzadan,

It is with great enthusiasm that we submit the final report of our capstone project, titled “AI Lecturer: Automated Course Content and Lecture Video Generation.” This report represents the culmination of months of collaboration, research, and innovation by our team to develop a system that transforms topic inputs into comprehensive educational resources.

The report provides a detailed overview of the project’s objectives, design methodology, system architecture, and implementation. Particular attention is given to the integration of AI technologies—including GPT-4, gTTS, and MoviePy—to automate the creation of slides, narration, quizzes, and video lectures. The system also features an intelligent Q&A chatbot designed to enhance student interaction.

Our goal with this project is to reduce the workload for educators and make high-quality, personalized learning content more accessible to students and institutions. We hope the insights presented in this document will demonstrate the potential of this tool in revolutionizing how digital lectures are created and delivered.

We are sincerely grateful for the opportunity to share our work with you and would greatly appreciate your feedback. Your input will be instrumental in refining the system for real-world deployment and broader educational impact. Should you have any questions or require further information, please feel free to reach out to any member of our team (Susmitha, Srikanth, Suryateja, Mounika, Vedanth, Anji, or Hima).

Thank you for your time, support, and thoughtful consideration.

Respectfully,

The AI Lecturer Capstone Team

EXECUTIVE SUMMARY

The **AI Lecturer** is an innovative educational platform developed to automate the creation of digital learning content, including lecture slides, narrated videos, and topic-based quizzes. Designed with both educators and students in mind, the system streamlines the traditionally time-consuming process of lecture preparation by leveraging the power of artificial intelligence and multimedia technologies.

The application offers four primary functions:

1. **Automated Slide Generation:** Using OpenAI's GPT-4, the system generates clear, structured, and well-organized slide content from a simple topic input. The slides are built using Reveal.js, ensuring a visually engaging presentation format.
2. **Narration and Video Lecture Creation:** The generated slide content is converted into audio using gTTS, with special attention given to mathematical expressions using MathReader-style prompting. Slide screenshots are then combined with audio narration using MoviePy to produce high-quality MP4 lecture videos.
3. **Adaptive Quiz Generation:** The system also generates multiple-choice quiz questions that align directly with the lecture content. These are formatted in JSON and rendered through an interactive frontend for student practice and self-assessment.
4. **GPT-4-Powered Chatbot for Q&A:** An integrated chatbot allows students to ask follow-up questions after watching the lecture. Using the original lecture content as context, GPT-4 provides accurate, personalized answers in real time.

This project addresses a growing need in modern education for scalable, accessible, and personalized teaching tools. By automating the entire pipeline—from content generation to assessment and interaction—the AI Lecturer empowers educators to save time and focus on higher-level teaching activities, while giving students an engaging and interactive learning experience.

This report provides a detailed overview of the system's objectives, design methodology, technical components, and implementation process. The AI Lecturer serves as a valuable asset for institutions, instructors, and learners seeking an efficient and intelligent solution for educational content delivery.

ACKNOWLEDGEMENTS

First and foremost, we would like to express our heartfelt gratitude to Dr. Behzadan and Dr. Sarraf for their continuous support, guidance, and encouragement throughout the development of the AI Lecturer project. Their valuable insights, timely feedback, and mentorship played a crucial role in shaping our ideas and refining the functionality of the system. Their dedication to academic excellence has been both inspiring and instrumental in the successful completion of this project.

We would also like to acknowledge the immense value of the open-source tools and AI technologies that formed the foundation of our work. The integration of platforms such as OpenAI GPT-4, Google Text-to-Speech (gTTS), Reveal.js, Playwright, MoviePy, and FastAPI enabled us to automate the content creation process and bring this project to life. These tools provided us with the flexibility and power needed to build an innovative and scalable solution for modern education.

We are also grateful to the University of New Haven for providing the infrastructure, resources, and academic environment that allowed us to explore, experiment, and grow as a team. The support of the faculty and the collaborative learning culture have greatly enriched our experience.

Lastly, we extend our sincere thanks to every member of our project team for their dedication, creativity, and teamwork. From conceptual design to system implementation, every step of this journey was made possible by the shared effort and enthusiasm of the team. This project stands as a testament to what can be achieved through collaboration and shared vision.

ABSTRACT

The AI Lecturer project presents a novel, fully automated system that generates educational lectures and adaptive assessments from a simple topic input—eliminating the need for manual content creation. Designed as a modular pipeline, the system seamlessly integrates state-of-the-art AI models and multimedia tools to produce narrated lecture videos and quizzes in under three minutes.

At the core of the system is **OpenAI’s GPT-4**, which generates structured slide content, explanatory narration, and multiple-choice questions. To address the limitations of conventional text-to-speech (TTS) in pronouncing mathematical expressions, the system implements a prompt-based transformation approach that rewrites formulas into natural, speech-friendly language. These rewritten texts are then synthesized into high-quality audio using **gTTS**.

Visual slides are dynamically rendered using **Reveal.js** and captured via **Playwright**. The slide images and narration clips are synchronized with **MoviePy** to produce a cohesive MP4 lecture video. Quizzes are concurrently generated using GPT-4 and rendered through a user-friendly interface.

The system supports dual deployment modes: a **Gradio prototype** for rapid prototyping and a **production-ready web application** built with **React and FastAPI** for scalable access. Each component—from content generation to video rendering—is modular, making the system adaptable for future enhancements such as avatar-based narration, retrieval-augmented grounding, or LMS integration.

AI Lecturer demonstrates how artificial intelligence can streamline the creation of personalized, accessible, and engaging educational experiences. By significantly reducing preparation time for educators while maintaining quality and interactivity, it redefines the possibilities of modern digital learning.

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1. INTRODUCTION

AI Lecturer

What Is AI Lecturer?

The **AI Lecturer** is an intelligent educational assistant that helps teachers, students, and educational institutions by automatically creating complete digital lectures. It's like having a virtual teaching assistant that never sleeps, works instantly, and delivers consistent results.

In a traditional setting, teachers spend hours planning a lesson. They need to:

- Research the topic
- Break it down into slides
- Write bullet points
- Record their voice explaining the slides
- Edit the video
- And finally, prepare a quiz for students

This process is time-consuming, especially when managing multiple subjects or large groups of students.

The AI Lecturer changes everything.

You simply type in a topic—like “**Photosynthesis**,” “**Data Science**,” or “**Climate Change**”—and the system takes care of the rest. Here's what it does:

◆ 1. Slide Generation

The system first writes structured, easy-to-understand slides. These slides look like what a human teacher would create—organized with clear headings, bullet points, and explanations that build understanding step by step.

◆ 2. Narration

Next, the system reads the slide content aloud using a realistic computer voice. This makes it feel like a teacher is speaking, guiding the student through each slide. If the topic includes complex math, the system can even rewrite those formulas into speech-friendly phrases (e.g., “a squared plus b squared equals c squared”) so it's easier to follow.

◆ 3. Lecture Video

The slides and narration are then combined into a **video lecture**. The result is a smooth, professional-looking video that students can watch anytime, anywhere—just like watching a recorded online class.

◆ 4. Interactive Quiz

To help students check their understanding, the system also creates a **quiz** related to the topic. It includes multiple-choice questions with correct answers based on what was taught in the lecture. This makes it easy for learners to test themselves and for teachers to assess their progress.

Who Can Benefit From It?

Teachers

- Save hours of time every week
- Focus more on student interaction and feedback instead of repetitive content creation
- Get ready-made materials for multiple topics in minutes

Students

- Learn at their own pace by watching clear, video-based lectures
- Get both visual and spoken explanations to suit different learning styles
- Practice with auto-generated quizzes to reinforce learning

Schools, Colleges, and Training Centers

- Easily scale up content creation for online courses or blended learning
- Support teachers by automating routine tasks
- Offer students 24/7 access to high-quality learning content

Why It Matters

In today's world—especially after the rise of online learning—tools like the AI Lecturer help bridge the gap between traditional classrooms and digital education. It helps ensure that every learner, no matter where they are, can access well-structured, engaging, and easy-to-understand lessons.

Whether used in schools, colleges, or for corporate training, the AI Lecturer is a **smart, fast, and reliable solution** for modern teaching and learning

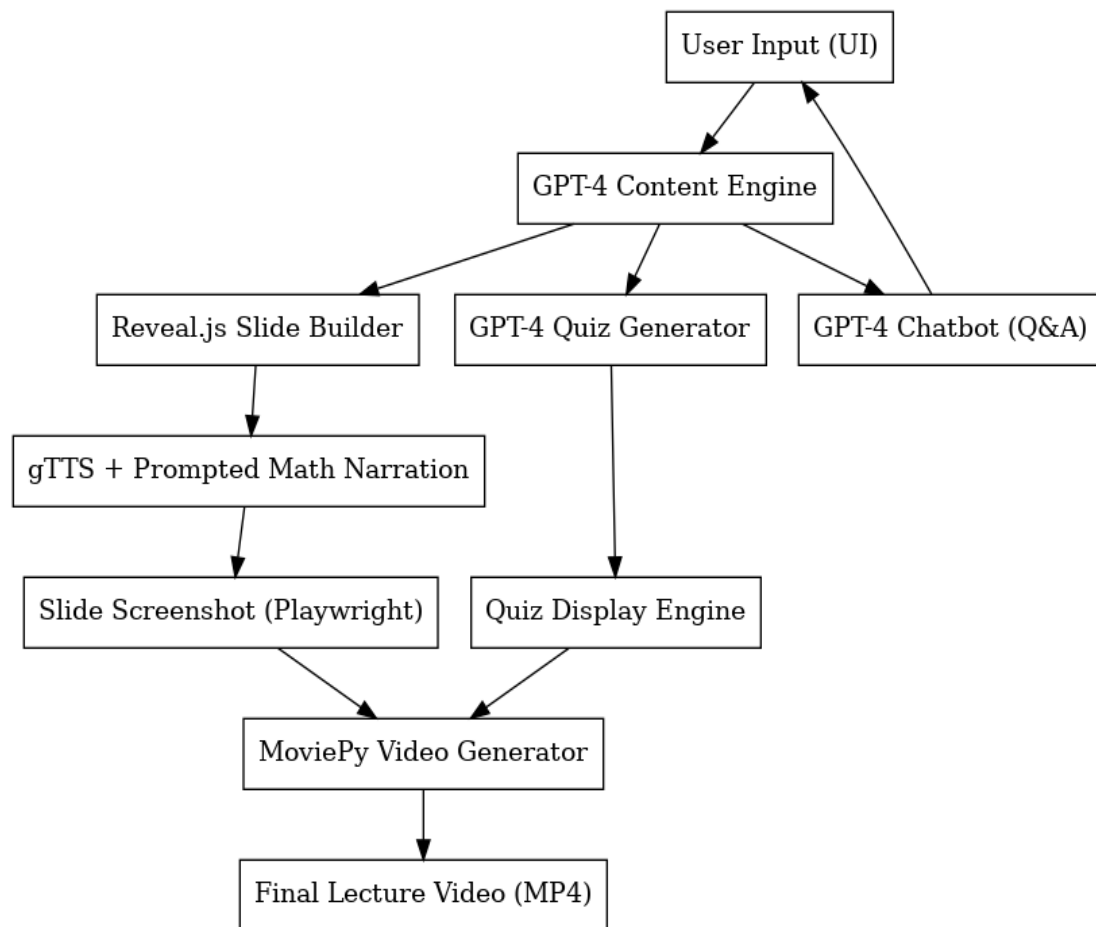
2. PROBLEM DEFINITION

The **AI Lecturer** is an AI-powered system designed to automatically generate educational content—including lecture slides, narrated video lectures, and interactive quizzes—from a simple topic input. The goal of the project is to reduce the manual effort and time involved in preparing digital learning materials by automating the entire content creation process using artificial intelligence and multimedia tools.

This system leverages GPT-4 for content generation, Google Text-to-Speech (gTTS) for audio narration, Reveal.js and Playwright for visual rendering, and MoviePy for video synthesis. It also includes a quiz generation feature and supports deployment via a user-friendly web interface using Gradio and React + FastAPI.

The AI Lecturer enables educators, students, and institutions to quickly create and deliver complete, engaging, and scalable learning experiences—making education more accessible, efficient, and personalized.

3. SYSTEM ARCHITECTURE



The **AI Lecturer** system is designed as a modular, automated content generation pipeline that transforms a single topic input into a complete digital lecture. This includes structured slides, AI-generated narration, video synthesis, and interactive quiz creation. The architecture emphasizes scalability, flexibility, and reusability by leveraging advanced AI models and multimedia tools in distinct stages.

Below is a detailed explanation of each component in the architecture:

1. User Input (UI):

The process begins with the user providing a topic through a user interface. This can be done through a **Gradio prototype interface** for quick testing or a **React.js-based web application** for production use. The topic serves as the core input around which the entire lecture is built.

2. GPT-4 Content Engine:

Once the topic is submitted, the system utilizes **OpenAI's GPT-4 API** to generate:

- Slide content (titles, bullet points, examples)
- Natural-language narration (including math descriptions)
- Quiz questions with multiple options and a correct answer

The GPT-4 engine functions as the knowledge processor and content designer, using carefully engineered prompts to simulate structured, educational output.

3. Reveal.js Slide Builder:

The structured slide content is formatted into a presentation using **Reveal.js**, an HTML-based slide framework. Each slide is encapsulated in `<section>` tags to ensure compatibility with visual rendering and navigation.

4. GPT-4 Quiz Generation:

In parallel, a secondary prompt to GPT-4 generates **multiple-choice questions** (MCQs) that directly reflect the lecture content. These quizzes are structured in JSON format and include the question, options, and the correct answer.

5. Narration Generation (gTTS + Prompted Math Narration):

To enhance accessibility, the **narration** for each slide is generated using **Google Text-to-Speech (gTTS)**. Before the text is passed to gTTS, mathematical expressions are rewritten by GPT-4 into speech-friendly descriptions—a technique referred to as **MathReader-style prompting**. This ensures clarity and proper pronunciation of mathematical formulas.

6. Slide Screenshot (Playwright):

Using **Playwright**, a headless browser automation tool, each slide is rendered and captured as a high-resolution image. This is done in a browser environment without a graphical interface to ensure consistency and speed.

7. Quiz Display Engine:

The previously generated quiz is parsed and rendered into an interactive format using either the Gradio or React interface. Students can take the quiz, receive instant feedback, and reinforce their understanding of the topic.

8. MoviePy Video Generator:

Slide images and corresponding audio narration are combined using **MoviePy**, a Python library for video editing. Each image is displayed for the duration of its respective audio narration, resulting in a coherent and fully narrated lecture video in MP4 format.

To ensure accurate timing, **Mutagen** is used to extract the duration of each audio clip.

9. Chatbot Q&A:

As an extension of the AI Lecturer system, the Chatbot module provides an interactive Q&A experience for students. It is designed to simulate a virtual teaching assistant, allowing learners to ask follow-up questions related to the generated lecture content.

Once the lecture slides and explanations are generated by GPT-4, the same content is passed as contextual input to a chat interface powered by GPT-4. Students can submit questions through the frontend (Gradio or React), and the system responds in real time with clear, context-aware answers.

This chatbot is particularly effective because it does not provide generic responses—instead, it uses the actual content of the generated lecture to ensure relevance and accuracy. The backend, built with FastAPI, manages communication between the user interface and the GPT-4 model.

10. Final Lecture Output:

The end product is a ready-to-use MP4 video lecture along with an interactive quiz. These can be downloaded, shared, or integrated into Learning Management Systems (LMS) for broader educational delivery.

COMPONENTS	TOOL/TECHNOLOGY USED
Topic Input	Gradio(prototype, React(production)
Content & Quiz Generation	OpenAI GPT-4 API
Slide Rendering	Reveal.js
Narration Synthesis	gTTS + GPT-(Math-Friendly narration)
Slide Screenshot Capture	Playwright (Headless Chromium)
Audio Duration Sync	Mutagen
Video Compilation	Moviepy
Quiz Interface	Gradio /React UI
Chatbot	GPT-4 API , FastAPI
Final Output	MP4 video+ JSON-formatted Quiz

4. TOOLS AND TECHNOLOGIES

Tools and Technologies

The AI Lecturer system combines cutting-edge AI, web development frameworks, and multimedia libraries to build an end-to-end lecture generation platform. Each tool plays a distinct role in the system's workflow, from content generation and slide rendering to narration, video creation, quiz delivery, and student interaction.

i. OpenAI GPT-4 API

- **Purpose:** AI-powered content generation.
- **Role in Project:** GPT-4 is the core engine behind most of the project's intelligence. It generates well-structured slides, natural language explanations, spoken-friendly narration content (including math rephrasing), and topic-specific multiple-choice quizzes. Additionally, it powers the chatbot module, which answers follow-up questions using the lecture content as context.
- **Why it's used:** GPT-4's language understanding and generation capabilities are unmatched in producing human-like educational content with minimal input

ii. Reveal.js

- **Purpose:** Slide rendering engine
- **Role in Project:** Converts GPT-4-generated content into clean, responsive HTML-based slides. Reveal.js supports navigation, themes, and animations, which enhance the visual presentation of lecture materials.
- **Why it's used:** It offers an elegant and customizable way to display content, ideal for converting plain text into professional-looking slides with minimal development overhead.

iii. Google Text-to-Speech (gTTS)

- **Purpose:** Converts text to audio narration
- **Role in Project:** Takes narration text from GPT-4 and generates corresponding MP3 audio files for each slide. This audio is later synchronized with slides to produce the final lecture video.
- **Why it's used:** gTTS offers simple, cloud-free speech synthesis that supports multiple languages and produces clear and intelligible output—perfect for voice-over generation in educational content.

iv. MathReader-Style Prompting (via GPT-4)

- **Purpose:** Ensures math readability in speech

- **Role in Project:** GPT-4 reformats mathematical expressions into natural spoken equivalents before sending them to gTTS. For example, $E = mc^2$ becomes “E equals m times c squared.”
- **Why it’s used:** This ensures that complex equations are understandable when spoken and avoids TTS mispronunciations, improving clarity for learners.

v. Playwright

- **Purpose:** Automated slide rendering and screenshot capture
- **Role in Project:** Opens Reveal.js slides in a headless browser and captures high-quality screenshots for each one. These images are essential for creating the video lecture.
- **Why it’s used:** Playwright is fast, reliable, and consistent across browsers, making it ideal for automating the visual rendering process.

vi. MoviePy

- **Purpose:** Video synthesis and editing
- **Role in Project:** Combines the captured slide images and the narration audio clips into a complete MP4 video. It handles slide timing, audio syncing, and export.
- **Why it’s used:** MoviePy is a powerful Python-based video editor that integrates smoothly with the AI pipeline for fully automated video production.

vii. Mutagen

- **Purpose:** Audio metadata handling
- **Role in Project:** Reads MP3 metadata (especially duration) to calculate how long each slide should be displayed in the final video.
- **Why it’s used:** Ensures precise timing between visuals and narration, improving video quality and synchronization.

viii. Gradio

- **Purpose:** Web-based prototype interface
- **Role in Project:** Provides a quick and easy frontend for users to input topics, preview slides and video, and take quizzes. Used mostly during development and demonstration phases.
- **Why it’s used:** Gradio allows rapid prototyping without needing a full frontend codebase—perfect for testing AI pipelines and user interaction.

ix. FastAPI

- **Purpose:** Backend API for data processing and routing
- **Role in Project:** Acts as the backend server for the AI Lecturer system, handling GPT-4 calls, TTS processing, video rendering, and communication between frontend and backend.

- **Why it's used:** FastAPI is lightweight, fast, and Python-based, making it perfect for serving machine learning workloads through RESTful APIs.

x. React.js

- **Purpose:** Frontend framework for deployment
- **Role in Project:** Used to build the full-featured user interface for topic submission, lecture playback, chatbot interaction, and quiz participation.
- **Why it's used:** React is modern, efficient, and scalable—making it the best choice for developing a responsive and dynamic learning interface.

xi. JSON (JavaScript Object Notation)

- **Purpose:** Structured data exchange
- **Role in Project:** Stores and transfers formatted data like quiz questions and slide content between components (GPT-4 output → backend → frontend).
- **Why it's used:** JSON is lightweight, easy to read, and well-supported across all systems, making it ideal for handling structured content.

Tool/Technology	Purpose	Role in Project
GPT-4 API	Content and quiz generation	Generates slides, narration text, quiz, and chatbot responses
Reveal.js	Slide rendering	Converts text into interactive HTML slides
gTTS	Text-to-speech synthesis	Produces narration audio for each slide
MathReader Prompting	Math speech enhancement	Converts math into readable phrases for audio
Playwright	Slide screenshot capture	Renders and screenshots each slide for video
MoviePy	Video editing	Combines images + audio into MP4 lecture video
Gradio	Prototype user interface	Enables quick testing and interaction with system
FastAPI	Backend server	Handles communication between frontend and processing layers
React.js	Production frontend	Builds the user-facing interface for deployed application
JSON	Data structure and exchange	Transfers quiz and content data between system components